



HITEC UNIVERSITY TAXILA CANTT
BS CYBER SECURITY
FALL 2024
AI LAB REPORT:3

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QUESTION NO:1

Write a Python function to calculate the factorial of a number (a non- negative integer). The function accepts the number as an argument.

CODE:

```
def factorial(num):  
    if num < 0:  
        return "Factorial not defined for negative numbers."  
    elif num == 0 or num == 1:  
        return 1  
    else:  
        fact = 1  
        for i in range(1, num + 1):  
            fact *= i  
        return fact  
  
n = int(input("Enter a number: "))  
print("Factorial of", n, "is:", factorial(n))
```

OUTPUT:

```
PS C:\Users\HP\Documents\New folder> &  
python.debugpy-2025.14.1-win32-x64\bund  
Enter a number: 6  
Factorial of 6 is: 720  
PS C:\Users\HP\Documents\New folder> █
```

QUESTION NO:2

Write a Python function that takes a list and returns a new list with unique elements of the first list.

a) Sample List : [1,2,3,3,4,4,5,3,7,5]

b) Unique List: [1, 2, 3, 4, 5, 7]

CODE:

```
def unique_list(lst):  
    return list(set(lst))  
  
sample_list = [1, 2, 3, 3, 4, 4, 5, 3, 7, 5]  
print("Original List:", sample_list)  
print("Unique List:", unique_list(sample_list))
```

OUTPUT:

```
PS C:\Users\HP\Documents\New folder> & 'c:\Use  
python.debugpy-2025.14.1-win32-x64\bundled\libs  
Original List: [1, 2, 3, 3, 4, 4, 5, 3, 7, 5]  
Unique List: [1, 2, 3, 4, 5, 7]  
PS C:\Users\HP\Documents\New folder>
```

QUESTION NO:3

Write a Python function to show the first 'n' even numbers and then return the sum and product of that even numbers outside the function.

CODE:

```
def even_numbers(n):  
    evens = []  
    for i in range(2, 2*n + 1, 2):  
        evens.append(i)  
    print("First", n, "even numbers:", evens)  
    return evens  
  
n = int(input("Enter how many even numbers you want: "))  
evens = even_numbers(n)  
sum_even = sum(evens)  
product_even = 1  
for i in evens:  
    product_even *= i  
  
print("Sum of even numbers:", sum_even)  
print("Product of even numbers:", product_even)
```

output:

```
Enter how many even numbers you want: 8
First 8 even numbers: [2, 4, 6, 8, 10, 12, 14, 16]
Sum of even numbers: 72
Product of even numbers: 10321920
PS C:\Users\HP\Documents\New folder> 
```

QUESTION NO:4

Write a python program to match two string using functions.

CODE:

```
def match_strings(str1, str2):
    if str1 == str2:
        print("Strings match!")
    else:
        print("Strings do not match.")

# Example:
s1 = input("Enter first string: ")
s2 = input("Enter second string: ")
match_strings(s1, s2)
```

OUTPUT:

```
python.debugpy-2025.14.1-win32-x64\bundled\libs\debugpy\
Enter first string: EVERY FAILURE TEACHES SOMETHING
Enter second string: DREAMS DONOT WORK UNLESS YOU DO
Strings do not match.
PS C:\Users\HP\Documents\New folder> 
```

QUESTION NO:5

Write a python program to name user login system using function.

CODE:

```
def login_system(username, password):  
    # Predefined credentials  
    correct_username = "HALIMA"  
    correct_password = "2005"  
  
    if username == correct_username and password == correct_password:  
        print("====Login successful! Welcome,====", username)  
    else:  
        print("Invalid username or password.")  
  
# Example:  
user = input("Enter username: ")  
pwd = input("Enter password: ")  
login_system(user, pwd)
```

OUTPUT:

```
Enter username: HALIMA  
Enter password: 567  
Invalid username or password.  
PS C:\Users\HP\Documents\New folder> |
```

CONCLUSION:

In this lab, we practiced creating and using functions in Python.

We developed programs for calculating factorials, finding unique elements in a list, generating even numbers with their sum and product, matching strings, and a simple login system.

This lab helped us understand the importance of functions for code reusability, simplicity, and structured programming.